Snowflake Project – Data Engineering & SQL Solution

Date	28 October 2026
Team ID	[Insert Your Actual Team ID, e.g., TM-1042]
Project Name	Health360: Comprehensive Healthcare Risk & Disease Analytics
Maximum Marks	5 Marks

This section evaluates the quality and completeness of the implemented Snowflake solution. The submission should demonstrate that the data has been prepared, cleaned, transformed, validated, and made ready for downstream analysis..

Instructions

1. Ensure SQL scripts are well-structured, clearly organized, and follow appropriate SQL practices.
2. The solution should address the project requirements through the required data preparation and transformation steps.
3. Include all SQL scripts, table/view definitions, and setup information needed to reproduce the solution.
4. Provide evidence that the final Snowflake data is accurate, consistent, and ready for analysis.

Solution Summary

Field	Details
Project / Repository Link	https://github.com/HarshsS777/Health360-Healthcare-Risk-Analytics
Data Source(s) Link	smart_healthcare_dataset.csv (5,000 anonymized patient records containing clinical vitals, demographics, and lifestyle factors).
Database	HEALTH360_DB
Warehouse	HEALTH360_WH (Virtual Warehouse, Standard X-Small).
Tables / Views Created	• RAW_PATIENT_DATA (Base staging table) • CLEAN_PATIENT_VITALS (Transformed &

	cleaned table) • VW_PATIENT_RISK_TIERS (Analytical view for High, Moderate, Low Risk Tiers) • VW_SYMPTOM_BURDEN (Analytical view for co-occurring conditions correlation)
Key Transformations / Calculations	• Age Stratification (18–34, 35–50, 51–65, 65+). • WHO BMI Categorization (Underweight, Normal Weight, Overweight, Obese). • Dynamic Risk Tier Assignment based on Systolic BP (≥ 140 or Glucose ≥ 140 = High Risk; BP ≥ 120 or Glucose ≥ 100 = Moderate Risk; else Low Risk).
KPIs created	• Total patients evaluated: 5,000; average blood pressure 134 mmHg; average glucose 134 mg/dL; average Dynamic Risk Tier Index 64. • Risk tier distribution: 3,479 High, 1,123 Moderate, 398 Low. • Dashboards: 2 master dashboards, 9 worksheets and a 4-scene Executive Story.
Key Insights	• 3,215 patients (64.3%) have no chronic condition; 1,207 (24.1%) have co-occurring Heart Disease & Stroke. • Patients with Heart Disease & Stroke are concentrated at 1–2 concurrent symptoms (440 and 465 patients). • Average blood pressure is nearly flat across exercise levels (133.8, 135.4, 134.1 mmHg). • Smoking and alcohol groups show higher Dynamic Risk Tier Index values on the lifestyle dashboard.

Code & SQL Implementation

The following scripts are executed in order in the HEALTH360_DB.PUBLIC_ANALYTICS schema using the HEALTH360_WH warehouse. Row counts and results noted in comments were reproduced from the 5,000-record source file.

Step 1 – Setup DDL: warehouse, database, schema, stage and RAW_PATIENT_DATA

```
-- 01_setup.sql
USE ROLE SYSADMIN;

CREATE WAREHOUSE IF NOT EXISTS HEALTH360_WH
  WAREHOUSE_SIZE      = 'XSMALL'
  AUTO_SUSPEND        = 60
  AUTO_RESUME         = TRUE
  INITIALLY_SUSPENDED = TRUE;

CREATE DATABASE IF NOT EXISTS HEALTH360_DB;
CREATE SCHEMA   IF NOT EXISTS HEALTH360_DB.PUBLIC_ANALYTICS;

USE WAREHOUSE HEALTH360_WH;
USE SCHEMA HEALTH360_DB.PUBLIC_ANALYTICS;

CREATE OR REPLACE FILE FORMAT FF_HEALTH_CSV
  TYPE = CSV
  FIELD_DELIMITER = ','
  SKIP_HEADER = 1
  FIELD_OPTIONALLY_ENCLOSED_BY = '"'
  NULL_IF = ('', 'NULL');

CREATE OR REPLACE STAGE STG_HEALTH360 FILE_FORMAT = FF_HEALTH_CSV;

-- Upload the CSV (SnowSQL) or use Snowsight > Add Data > Load files:
-- PUT file://smart_healthcare_dataset.csv @STG_HEALTH360 AUTO_COMPRESS = TRUE;

CREATE OR REPLACE TABLE RAW_PATIENT_DATA (
  age               INTEGER,
  gender            VARCHAR(10),
  bmi               FLOAT,
  exercise_level    INTEGER,
  smoking           INTEGER,
  alcohol           INTEGER,
  blood_pressure    INTEGER,
  cholesterol       INTEGER,
  glucose           INTEGER,
  fatigue           INTEGER,
  chest_pain        INTEGER,
  dizziness         INTEGER,
  heart_disease     INTEGER,
  diabetes          INTEGER,
  stroke            INTEGER,
  health_risk_score FLOAT
);

COPY INTO RAW_PATIENT_DATA
  FROM @STG_HEALTH360
  FILE_FORMAT = (FORMAT_NAME = FF_HEALTH_CSV)
  ON_ERROR = 'ABORT_STATEMENT';

SELECT COUNT(*) AS raw_rows FROM RAW_PATIENT_DATA;   -- expected: 5000
```

Step 2 – Data cleaning and feature engineering: CLEAN_PATIENT_VITALS

```
-- 02_clean_transform.sql
CREATE OR REPLACE TABLE CLEAN_PATIENT_VITALS AS
WITH validated AS (
  SELECT
    age,
    INITCAP(TRIM(gender)) AS gender,
    bmi, exercise_level, smoking, alcohol,
    blood_pressure, cholesterol, glucose,
    fatigue, chest_pain, dizziness,
    heart_disease, diabetes, stroke
  FROM RAW_PATIENT_DATA
  WHERE age            BETWEEN 0 AND 120
    AND bmi            BETWEEN 10 AND 70
    AND blood_pressure BETWEEN 60 AND 260
    AND cholesterol    BETWEEN 80 AND 500
```

```
    AND glucose        BETWEEN 40 AND 600
    AND exercise_level IN (0, 1, 2)
    AND smoking IN (0, 1) AND alcohol IN (0, 1)
    AND fatigue IN (0, 1) AND chest_pain IN (0, 1) AND dizziness IN (0, 1)
    AND heart_disease IN (0, 1) AND diabetes IN (0, 1) AND stroke IN (0, 1)
    AND gender IS NOT NULL
)
SELECT
  ROW_NUMBER() OVER (ORDER BY age, gender, bmi, blood_pressure,
                            glucose, cholesterol)        AS patient_id,
  v.*,
  -- Age stratification
  CASE WHEN age <= 34 THEN '18-34 (Young Adult)'
       WHEN age <= 50 THEN '35-50 (Middle Aged)'
       WHEN age <= 65 THEN '51-65 (Senior)'
       ELSE                '65+ (Elderly)' END             AS age_bracket,
  -- WHO BMI categories
  CASE WHEN bmi < 18.5 THEN 'Underweight'
       WHEN bmi < 25   THEN 'Normal Weight'
       WHEN bmi < 30   THEN 'Overweight'
       ELSE                 'Obese' END                    AS bmi_category,
  -- Dynamic risk tier (systolic BP / glucose rule)
  CASE WHEN blood_pressure >= 140 OR glucose >= 140 THEN 'High Risk'
       WHEN blood_pressure >= 120 OR glucose >= 100 THEN 'Moderate Risk'
       ELSE                                             'Low Risk' END
                                                          AS risk_tier,
  -- Symptom burden (0-3) and primary condition
  fatigue + chest_pain + dizziness                        AS symptom_burden_count,
  CASE WHEN heart_disease = 1 AND stroke = 1 THEN 'Heart Disease & Stroke'
       WHEN heart_disease = 1                THEN 'Heart Disease'
       WHEN diabetes = 1                     THEN 'Diabetes'
       WHEN stroke = 1                       THEN 'Stroke'
       ELSE                                       'No Chronic Condition' END
                                                          AS primary_disease_condition
FROM validated v;
```

Step 3 – Analytical views: VW_PATIENT_RISK_TIERS and VW_SYMPTOM_BURDEN

```
-- 03_views.sql
CREATE OR REPLACE VIEW VW_PATIENT_RISK_TIERS AS
SELECT
  risk_tier,
  exercise_level,
  gender,
  COUNT(*)                        AS patient_count,
  ROUND(AVG(blood_pressure), 1)   AS avg_blood_pressure,
  ROUND(AVG(cholesterol), 1)      AS avg_cholesterol,
  ROUND(AVG(glucose), 1)          AS avg_glucose
FROM CLEAN_PATIENT_VITALS
GROUP BY risk_tier, exercise_level, gender;

CREATE OR REPLACE VIEW VW_SYMPTOM_BURDEN AS
SELECT
  primary_disease_condition,
  symptom_burden_count,
  COUNT(*)                        AS patient_count
FROM CLEAN_PATIENT_VITALS
GROUP BY primary_disease_condition, symptom_burden_count;
```

Step 4 – Analytical queries: age stratification, BMI distribution, lifestyle impact

```
-- 04_analysis.sql
-- 4.1 Cohort age stratification x WHO BMI category
SELECT bmi_category, age_bracket, COUNT(*) AS patients
FROM CLEAN_PATIENT_VITALS
GROUP BY bmi_category, age_bracket
ORDER BY bmi_category, age_bracket;
-- Normal Weight row: 552 | 524 | 485 | 449  (18-34, 35-50, 51-65, 65+)

-- 4.2 BMI distribution (share of cohort)
SELECT bmi_category,
       COUNT(*)                                          AS patients,
       ROUND(100 * COUNT(*) / SUM(COUNT(*)) OVER (), 1)  AS pct_of_cohort
FROM CLEAN_PATIENT_VITALS
GROUP BY bmi_category
ORDER BY patients DESC;

-- 4.3 Lifestyle risk factors (smoking x alcohol)
SELECT smoking, alcohol,
       COUNT(*)                                          AS patients,
       ROUND(AVG(blood_pressure), 1)                     AS avg_bp,
       ROUND(AVG(glucose), 1)                            AS avg_glucose,
       ROUND(100 * AVG(IFF(risk_tier = 'High Risk', 1, 0)), 1)
                                                         AS pct_high_risk
FROM CLEAN_PATIENT_VITALS
```

```
GROUP BY smoking, alcohol
ORDER BY smoking, alcohol;

-- 4.4 Exercise level vs vitals
SELECT exercise_level,
       ROUND(AVG(blood_pressure), 3) AS avg_bp,
       ROUND(AVG(glucose), 1)        AS avg_glucose
FROM CLEAN_PATIENT_VITALS
GROUP BY exercise_level
ORDER BY exercise_level;
-- avg_bp: 133.838 | 135.437 | 134.050  (exercise levels 0, 1, 2)

-- 4.5 Symptom burden heatmap source
SELECT * FROM VW_SYMPTOM_BURDEN
ORDER BY primary_disease_condition, symptom_burden_count;
```

Step 5 – Validation checks

```
-- 05_validation.sql
SELECT (SELECT COUNT(*) FROM RAW_PATIENT_DATA)      AS raw_rows,    -- 5000
       (SELECT COUNT(*) FROM CLEAN_PATIENT_VITALS)  AS clean_rows;  -- 5000

SELECT risk_tier, COUNT(*) AS patients
FROM CLEAN_PATIENT_VITALS GROUP BY risk_tier;
-- High Risk 3479 | Moderate Risk 1123 | Low Risk 398

SELECT COUNT(*) AS null_vitals
FROM CLEAN_PATIENT_VITALS
WHERE age IS NULL OR bmi IS NULL OR blood_pressure IS NULL
   OR cholesterol IS NULL OR glucose IS NULL;               -- expected: 0

SELECT ROUND(AVG(blood_pressure), 1) AS avg_bp,             -- 134.4
       ROUND(AVG(glucose), 1)        AS avg_glucose         -- 134.5
FROM CLEAN_PATIENT_VITALS;
```

Snowflake Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Required data is successfully loaded and accessible in Snowflake	
2	Data is cleaned, transformed, and prepared appropriately	
3	Required tables, views, and data relationships are correctly implemented	
4	Required SQL logic, calculations, and KPIs are correctly implemented	
5	Data and solution outputs are validated against the project requirements	
6	Solution is properly organized, documented, and functional	

Additional Notes / Comments:

Figure 1: Health360 Snowflake Data Engineering Execution & Web Deployment Shell

